Field Hours

CSCE 645 Geometric Modeling

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Four points and de Casteljau

You build a cubic Bezier curve from 4 control points. It starts at the first point and ends at the last. The middle two pull the shape but the curve need not pass through them. At t = 0 you are at the first point, and at t = 1 you are at the last. You evaluate t by repeated linear blends, the de Casteljau step, instead of expanding a polynomial.

You take P0 = (0, 0), P1 = (1, 2), P2 = (2, 2), and P3 = (3, 0). At t = 0.5 each blend is a midpoint, so the first step yields (0.5, 1), (1.5, 2), and (2.5, 1). Two more steps land on the curve. You keep all 4 points, because moving P1 changes the leaving tangent at P0 and does not move the endpoint P0 itself.

de Casteljau evaluates a cubic by blending, not by expanding powers of t.

1. Start with 4 points

2. Blend each pair at t

3. Blend the results

4. Land on the curve

The sample cubic at t = 0.5

1. The first blends are the midpoints (0.5, 1), (1.5, 2), and (2.5, 1).

2. The second blends are the midpoints of those, (1, 1.5) and (2, 1.5).

3. The third blend is (1.5, 1.5). That point lies on the curve. Parameter 0.5 is not a claim about arc length.

Problem 1. At t = 0, which control point does the cubic pass through?

Answer. P0, the first control point. The curve passes through the first and last of the 4 points, not through the middle two.

Problem 2. You move P1 from (1, 2) to (1, 4). Does P0 move?

Answer. No. P0 stays at (0, 0). Moving P1 changes the tangent that leaves P0.

Lecture 2 · 90 minutes

Joins that match, or do not

You join two cubic segments so a longer wire can look like one curve. Position continuity, called C0, means the end of the first segment and the start of the second are the same point. If both are (3, 0), the join has no gap. You can still have a corner, because C0 says nothing about direction. The mechanism is a shared point, not a shared

tangent.

You get a C1 join when the outgoing tangent of the first segment matches the incoming tangent of the second in direction and length. With the shared point at (3, 0), a shared tangent vector of (1, -1) makes the join C1. Matching direction only, while the second tangent is twice as long, is geometric smoothness, G1. You use G1 when the parameter speed may jump.

A shared point closes a gap. A shared tangent vector removes the corner.

C0 join

Same endpoint

A corner is allowed

Tangents may differ

C1 join

Same endpoint

Same tangent vector

No corner at the join

Two segments at (3, 0)

1. Segment A ends at (3, 0) with tangent vector (1, -1).

2. Segment B starts at (3, 0), and its first control sits at (4, -1), the same offset of (1, -1).

3. Position and tangent both match, so the join is C1. If B started toward (4, 1), you would have C0 only.

Problem 1. Two segments meet at (3, 0) but their tangents point opposite ways. Which continuity do you have?

Answer. C0 only. The positions match, so there is no gap, but the tangent vectors do not match, so the join is not C1.

Problem 2. The tangents match in direction, but the second is twice as long. Is the join C1?

Answer. No. C1 needs the tangent vectors to match in direction and length. Matching direction only is G1.

Lecture 3 · 90 minutes

A spline that edits locally

You use a cubic B-spline when you want local control. A point on the curve depends on 4 consecutive control points, not on the whole chain. Move control point number 7 in a chain of 12 and only the nearby spans change. You do not rerun the blend on the distant spans. The 4 basis weights in a span sum to 1, so the curve stays in their convex hull.

You place the parameter on uniform knots spaced by 1, so span k runs from parameter k to k + 1. Inside one span

you blend the 4 active controls. At a knot of multiplicity 1 the cubic stays C2: position, tangent, and curvature agree. Raise that knot to multiplicity 3 and the curve can pass through a control and lose that smoothness. That is how you cut a corner without splitting the chain by hand.

A cubic span reads four controls. The other controls in the chain stay idle.

One span, four controls

Weights that sum to 1

Distant spans stay put

Editing point 7 of 12

1. The chain has 12 control points. You move point 7 from (7, 0) to (7, 2).

2. A cubic span uses 4 consecutive controls, so a span that does not include point 7 is unchanged.

3. You recompute only the spans that include point 7. The rest of the chain keeps its old points.

Problem 1. How many control points does one cubic B-spline span use?

Answer. 4. A point on a cubic span is a blend of 4 consecutive controls, and those basis weights sum to 1.

Problem 2. You raise one knot from multiplicity 1 to multiplicity 3. What agreement can you lose?

Answer. The C2 agreement at that knot. Multiplicity 1 keeps position, tangent, and curvature matched. Multiplicity 3 can put a corner on a control.

Lecture 4 · 90 minutes

Sixteen controls on a patch

You make a surface patch as a grid of curves. A bicubic Bezier patch has 4 rows of 4 control points, 16 points in all. You pick parameters u and v, each from 0 to 1. First you run de Casteljau along each row at the chosen u, which gives you 4 points. Then you run it down those 4 points at v. One pair (u, v) yields one point on the patch.

You read each boundary off the border of the grid. The edge at v = 0 is the cubic of the first row of 4 controls. A corner of the patch is a corner control, so the patch passes through its 4 corners and not through the other 12 controls. You move an interior control to bulge the middle without sliding a corner. The corner stays on its control point.

A bicubic patch is two nested curve evaluations on a 4 by 4 grid.

16 control points -> Four blends in u -> One blend in v -> A surface point

A flat patch, then a bulge

1. All 16 controls have z = 0. The corners are (0, 0, 0), (3, 0, 0), (0, 3, 0), and (3, 3, 0).

2. At (u, v) = (0.5, 0.5) every blend stays in the plane z = 0.

3. Raise the interior controls by 2 and evaluate again. The corners stay on z = 0 and the center rises.

Problem 1. How many control points does a bicubic Bezier patch have?

Answer. 16. The grid is 4 rows of 4 controls.

Problem 2. You move one interior control. Which of the 4 corners move?

Answer. None. The patch passes through its 4 corner controls. An interior control bulges the middle and does not slide a corner.

Lecture 5 · 90 minutes

A lattice of 64 points

You extend the patch by one parameter when the shape has thickness. A tricubic Bezier solid uses a lattice of 4 by 4 by 4 controls, which is 64 points. A parameter triple (u, v, w), each from 0 to 1, picks one point in the volume. You evaluate it as three nested de Casteljau passes, along u, then v, then w. The 8 corners of the lattice are corners the map must hit.

You use the solid for a bracket that is 2 units thick, with the outer boundary on one parameter face and the inner boundary on the opposite face. Deform the lattice and both boundaries follow, so thickness is not painted on later. You still test for a fold. If two different triples land on one point, the solid has pinched, and a later slice will lie to you about what is inside.

One more parameter turns a patch into a volume. A fold means two triples collide.

64 control points

Three nested blends

A point in the volume

Reject a folded map

Two corners and a center parameter

1. The lattice is 4 by 4 by 4. Parameter (0, 0, 0) returns the control at that corner.

2. Parameter (1, 1, 1) returns the opposite corner. Those two are among the 8 corners the solid passes through.

3. Parameter (0.5, 0.5, 0.5) is inside the parameter cube. It is a blend of controls, not automatically one center control.

Problem 1. How many control points are in a 4 by 4 by 4 tricubic lattice?

Answer. 64. You multiply 4 by 4 by 4.

Problem 2. Two different parameter triples evaluate to the same point. What is wrong?

Answer. The solid has folded or pinched. Different triples should land on different points, or a slice through the volume is no longer trustworthy.

Lecture 6 · 90 minutes

Vertices, edges, and faces

You describe a mesh by connectivity, not by where the vertices sit. Count vertices V, edges E, and faces F. One triangle has V = 3, E = 3, and F = 1. For a mesh that is a single disk, V minus E plus F equals 1, and 3 minus 3 plus 1 equals 1. You recompute that count after every edit. A wrong total means a hole, a pinch, or an edge you counted twice.

You weld a second triangle onto one edge. The new face adds 1 face, 2 edges, and 1 vertex, because the shared edge and its 2 vertices are already stored. The totals become V = 4, E = 5, and F = 2, and 4 minus 5 plus 2 is still 1. You store the shared edge once, with two face pointers. Storing it twice inflates E even though the picture looks welded.

The same disk count before and after a weld. The shared edge is stored once.

One triangle

V 3, E 3, F 1

Count equals 1

No shared edge

Two triangles welded

V 4, E 5, F 2

Count still equals 1

One edge, two faces

Weld along one edge

1. Start from V = 3, E = 3, F = 1. The disk count is 1.

2. Add a triangle on one edge: plus 1 vertex, plus 2 edges, plus 1 face.

3. The result is V = 4, E = 5, F = 2. Then 4 minus 5 plus 2 equals 1, so the mesh is still a disk.

Problem 1. A disk mesh has V = 6 and E = 8. What must F be so that V minus E plus F equals 1?

Answer. 3. Six minus 8 is -2, and -2 plus 3 equals 1.

Problem 2. You store the shared edge of the two-triangle mesh as two edges. What is E, and what is the count?

Answer. E becomes 6 instead of 5. Then V minus E plus F is 4 minus 6 plus 2, which equals 0, so the count no longer matches a disk.

Lecture 7 · 90 minutes

Union, intersection, difference

You combine two solids with a Boolean test instead of dragging vertices. Union keeps any point inside either

solid. Intersection keeps a point only when it is inside both. Difference keeps points inside the first and outside the second. You classify each candidate with 2 inside tests. A point inside both is in the intersection, and that same point is rejected by difference.

You apply the tests to a box and a cylinder. The box occupies x and y from 0 to 4, and z from 0 to 2. The cylinder has radius 1 and its axis along z through (2, 2). A point at (2, 2, 1) is inside both, so intersection keeps it and difference removes it. A point at (0.2, 0.2, 1) is inside the box only, so difference keeps it and intersection discards it.

Each Boolean is two inside tests and a rule for how the two answers combine.

1. Test the first solid
2. Test the second solid
3. Apply one rule
4. Keep or drop the point

Box minus cylinder

1. The box runs from 0 to 4 in x and y, and from 0 to 2 in z. The cylinder has radius 1 about (2, 2).
2. The point (2, 2, 1) is inside both solids, so difference rejects it.
3. The point (0.2, 0.2, 1) is inside the box and outside the cylinder, so difference keeps it.

Problem 1. A point is inside the box and inside the cylinder. Is it in the intersection?

Answer. Yes. Intersection keeps a point only when both inside tests pass, and both pass here.

Problem 2. That same point is tested for box minus cylinder. Do you keep it?

Answer. No. Difference keeps points inside the first solid and outside the second. This point is inside the cylinder, so it is removed.

Lecture 8 · 90 minutes

A model you can check

You can now lay a cubic through 4 controls with de Casteljau, join two cubics as C0 or C1, and edit one span of a 12-point B-spline without touching the far spans. You can fill a 16-point patch and a 64-point solid, check a disk mesh with V minus E plus F, and combine a box with a cylinder by union, intersection, or difference. Each construction has a count you can recompute.

You close a model by checking shape and connectivity together. A join that is only C0 still shows a corner even when the Boolean solid is closed. A mesh whose disk count is not 1 will break a later cut even if the Bezier boundary looked smooth. You run the tangent test at the join and the count on the mesh before you trust the solid you are about to hand off.

Build, then check the join and the count, then combine solids.

Build the curves -> Fill patch or solid -> Check V, E, and F -> Boolean combine

A disk under a smooth wire

1. Two cubics meet at (3, 0) with the same tangent vector, so the wire join is C1.

2. The mesh under the wire is the welded pair: V = 4, E = 5, F = 2, and the disk count is 1.

3. You then subtract a cylinder of radius 1. A point inside both solids is removed, and you recompute the count after the cut.

Problem 1. Combine continuity and topology. Two cubics meet at one point but their tangents differ, and the mesh has V = 4, E = 5, F = 2. What is true of each?

Answer. The mesh is still a disk because 4 minus 5 plus 2 equals 1, but the curve join is only C0. The corner stays until the tangent vectors match.

Problem 2. A bicubic patch has 16 controls and you move 1 interior control. How many corners stay fixed?

Answer. All 4 corners stay fixed. You rerun the nested de Casteljau steps for the (u, v) pairs you care about, because an interior control changes the interior only.